Phishing Email Detection and Awareness System

Cyber Security Internship – Task 2

Introduction

Phishing is one of the most common cyber attacks used by attackers to steal sensitive information such as passwords, banking credentials, and personal data. Attackers send fake emails that appear to come from trusted organizations such as banks, online services, or company IT teams.

These emails attempt to trick users into clicking malicious links, downloading harmful attachments, or providing confidential information. Many phishing attacks succeed because users are unaware of phishing techniques and warning signs.

This project focuses on analyzing phishing email examples, identifying phishing indicators, and creating awareness guidelines to help users recognize and avoid phishing attacks.

Objective

The objective of this task is to:

- Analyze phishing email samples
- Identify phishing indicators
- Classify email risk levels
- Explain how phishing attacks work
- Provide prevention and awareness guidelines

This task helps improve cybersecurity awareness and teaches users how to identify suspicious emails.

Tools Used

The following tools were used for phishing email analysis:

- Web Browser (for inspecting links and domains)

- Email Header Analyzer tools
- Documentation tools such as Google Docs / MS Word
- Public phishing email datasets for learning

Example email header analysis tools used for investigation:

- https://toolbox.googleapps.com/apps/messageheader/
- https://mxtoolbox.com/EmailHeaders.aspx

These tools help analyze sender information, domain authenticity, and email routing details.

Phishing Email Sample Analysis

Email Example 1 – Bank Account Verification Scam

Subject: Urgent: Verify Your Bank Account Immediately

Email Content

Dear Customer,

We detected unusual activity in your bank account.
Your account will be temporarily suspended unless you verify your details.

Click here to verify your account:
http://secure-bank-login-verification[.]com

Regards
Bank Security Team

Identified Indicators

Indicator	Explanation
Generic greeting	Email uses "Dear Customer" instead of real name
Urgent language	Creates panic to force quick action

Suspicious link	Unknown domain not related to the bank
Fake domain	Appears similar to official banking site

Risk Classification

Phishing Email

Attack Explanation

The attacker attempts to scare the user by claiming suspicious activity on their account. The email includes a verification link leading to a fake website designed to collect login credentials.

Email Example 2 – Fake Microsoft Password Reset

Subject: Microsoft Security Alert: Password Expiring Today

Email Content

Hello User,

Your Microsoft account password will expire today.
You must reset your password immediately to avoid losing access.

Reset Password:
http://microsoft-account-security-reset[.]net

Microsoft Security Team

Identified Indicators

Indicator	Explanation
Fake domain	Not an official Microsoft domain
Urgency	Forces user to act immediately

Generic greeting	Does not include the user’s name
Suspicious URL	External site not related to Microsoft

Risk Classification

Phishing Email

Attack Explanation

This email impersonates Microsoft security notifications. The link redirects users to a fake password reset page where attackers can steal login credentials.

Email Example 3 – Fake Package Delivery Notification

Subject: Delivery Failed – Update Shipping Information

Email Content

Dear Customer,

Your package delivery has failed due to an incomplete address.

Please update your shipping information within 24 hours.

Track your package:
http://delivery-update-center[.]info

Shipping Department

Identified Indicators

Indicator	Explanation
Unknown sender	Not associated with any real courier service
Suspicious link	Domain not related to delivery companies
Urgency	Forces user to act within 24 hours

Generic greeting	Does not include recipient name

Risk Classification

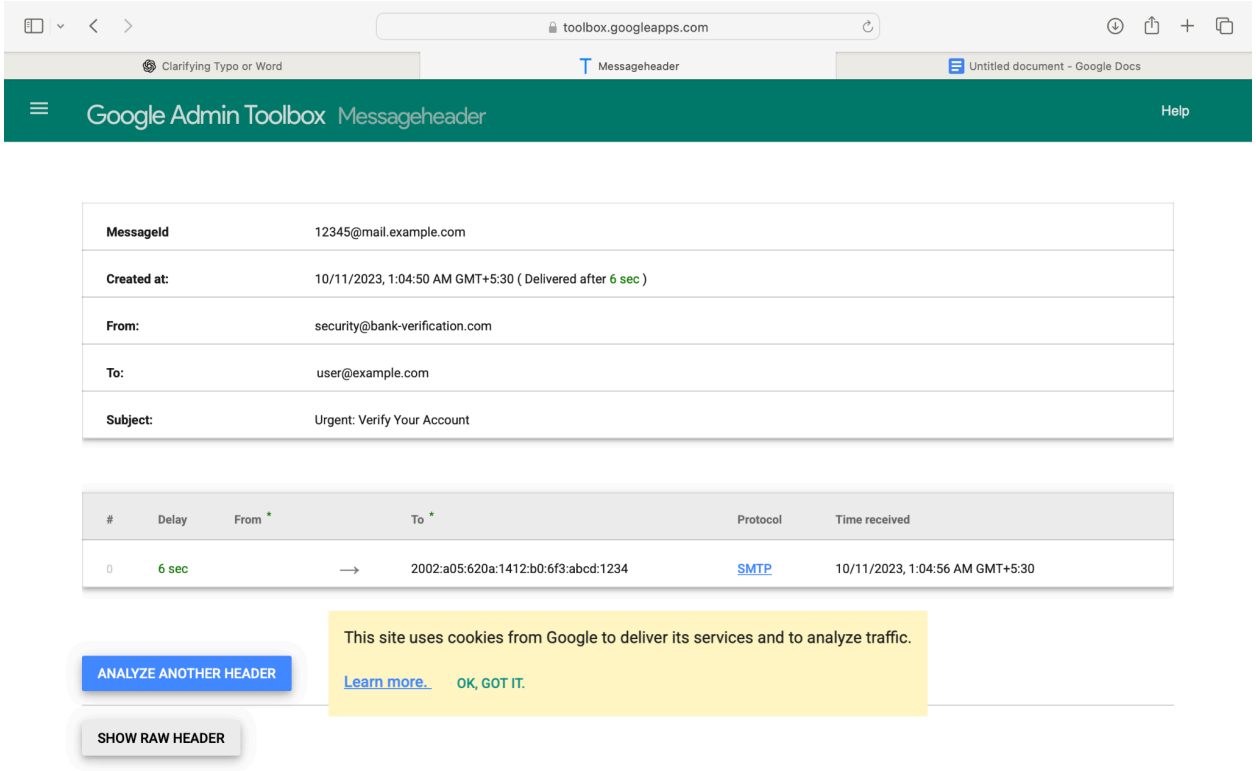
Phishing Email

Attack Explanation

This phishing email attempts to trick users into entering personal details on a fake delivery website. The attacker may collect addresses, phone numbers, or payment information.

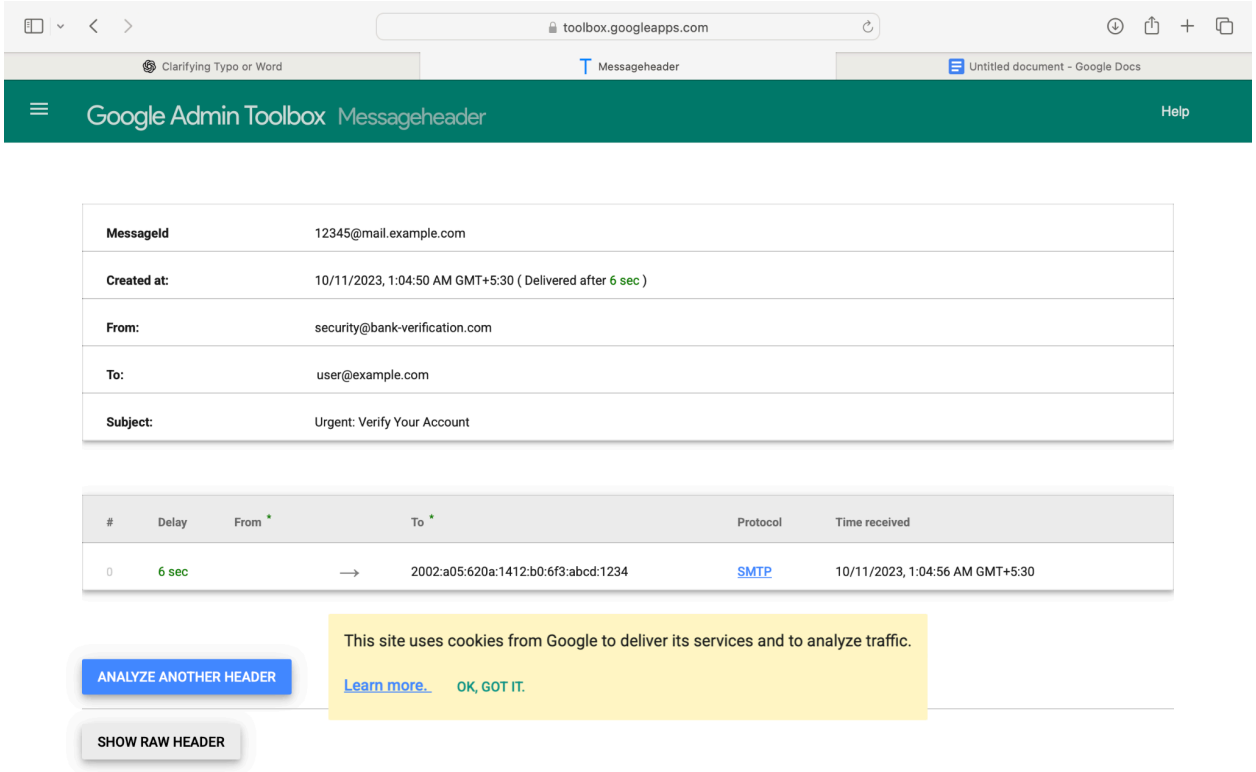
Images

Email 1 Bank Verification Phishing Email



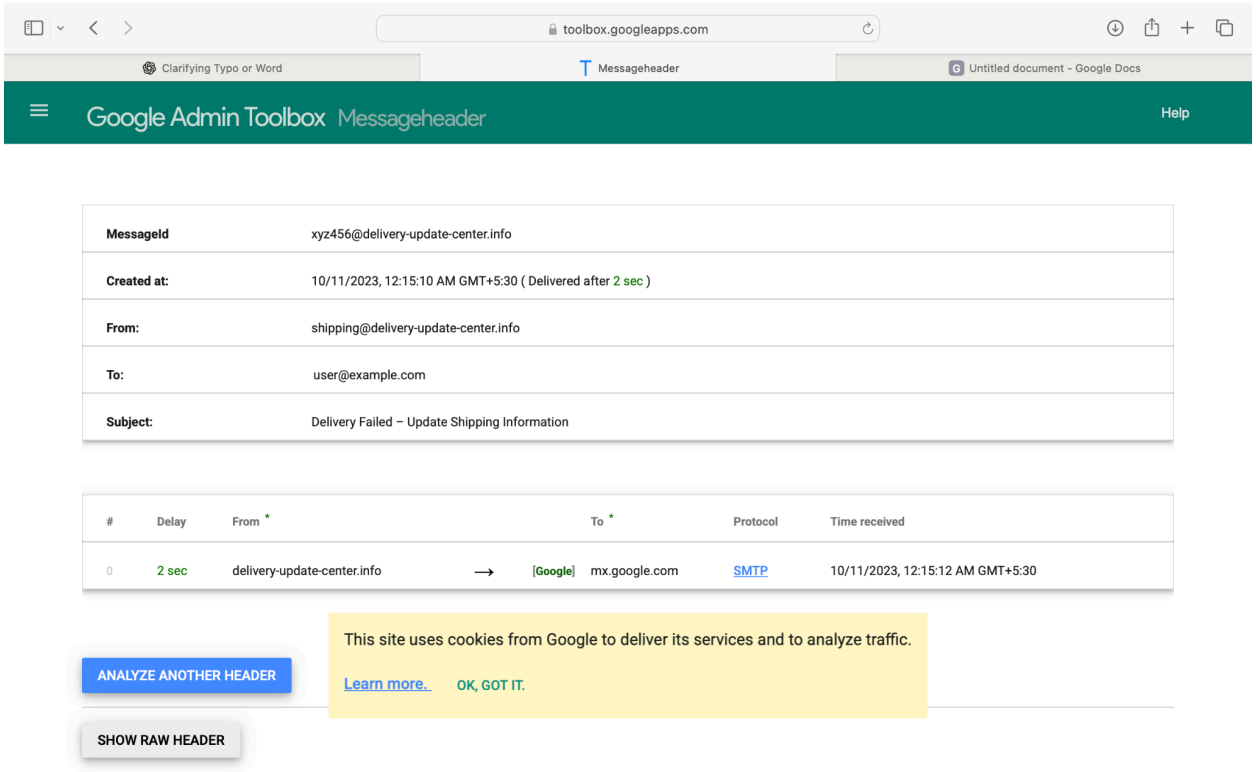
Message header analysis of a phishing email pretending to be from a bank.

Emai 2 Email 2 Microsoft Password Reset Phishing Email



Message header analysis of a phishing email impersonating Microsoft security alert

Email 3 Fake Delivery Notification Email



Message header analysis of a phishing email related to fake delivery notification.

Risk Classification Summary

Email Example	Sender Email	Subject	Suspicious Indicators	Risk Level
Email 1 – Bank Verification Scam	security@bank-verification.com	Urgent: Verify Your Account	Fake bank domain, urgent language, request to verify account	High Risk – Phishing

Email 2 – Delivery Scam	shipping@delivery-update-center.info	Delivery Failed – Update Shipping Information	Unknown sender domain, fake delivery notice, suspicious link	High Risk – Phishing
Email 3 – Microsoft Password Reset	security@microsoft-account-security-reset.net	Microsoft Security Alert: Password Expiring Today	Fake Microsoft domain, urgent password reset request	High Risk – Phishing

Indicators Identified

During the analysis, several common phishing indicators were identified:

- Fake or suspicious sender domains
- Urgent or threatening language
- Suspicious links leading to unknown websites
- Generic greetings instead of user names
- Requests for sensitive information

These indicators are commonly used by attackers to manipulate victims.

Prevention Guidelines

Users can protect themselves from phishing attacks by following these guidelines:

- Always verify the sender’s email address
- Avoid clicking suspicious links
- Check website URLs carefully before entering credentials
- Use multi-factor authentication for accounts
- Report suspicious emails to the organization’s security team

Following these practices can significantly reduce the risk of phishing attacks.

Employee Awareness Tips

Do’s

- Verify email sender before responding
- Hover over links before clicking them
- Contact the organization directly if unsure
- Report suspicious emails immediately

Don’ts

- Do not share passwords or OTPs via email
- Do not download attachments from unknown sources
- Do not click links from suspicious emails

User awareness is one of the most effective defenses against phishing.

Conclusion

Phishing attacks remain a major cybersecurity threat affecting individuals and organizations worldwide. Attackers exploit human psychology by creating urgency, impersonating trusted organizations, and providing malicious links.

Through the analysis of phishing email examples, this project identified common phishing indicators and demonstrated how attackers attempt to deceive users. Educating users about these indicators is essential to reducing the success rate of phishing attacks.

Organizations must continue to promote cybersecurity awareness and encourage employees to verify suspicious communications before taking action.